An Eye-to-Hand Visual Servoing Controller of 3-DOF Robot Using Feature Matching Technology

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***Abstract* – In robotic control sectors, the image-based visual servo method is an advanced approach used to control the robot to the desired position by using pixels from cameras. We can control the robot flexibly to achieve high quality when the joint variables have little backflash. Applications of this method can be classified into two groups: eye-in-hand and eye-to-hand ones. In this paper, we employ the eye-to-hand method for position tracking control problems of robotic manipulators. Kinematics of the robot and camera are first derived based on homogenous transformation relation and pinhole models. The control technique uses the normalized difference between the pixel observed from the camera and the desired pixel to result in proper control signals throughout a flexible activation gain. The feasibility of this method will be demonstrated through comparative simulation results.**

Keywords* – *eye to hand, camera, the error of pixel, flexible PBVS.

I. INTRODUCTION

Nowadays, vision control of robotic systems has been widely adopted in many applications from manipulators [1, 2], mobile robots [3], to underwater vehicles [4, 5]. Visual servoing technology is an interesting motion control approach that uses pixel measurement to control robot motions to desired positions. Such the technique is categorized into two classes: position-based visual servoing (PBVS) and image-based visual servoing (IBVS). Each type has different advantages and disadvantages [6, 7]. The PBVS methods need to obtain the geometric shape of the target, internal parameters of the camera and observable image plane characteristics. The main task of PBVS is to reduce the difference between the current shape and the desired shape. So, the calibration accuracy of the geometric model and camera of the target seriously affects and limits the performance of the PBVS. The IBVS methods are fundamentally different from the PBVS in that they do not estimate the relative pose of the target. Hence, they have been widely used in industrial sectors because of its robustness, such as: unmanned systems [8], aircrafts [9], fault diagnosis [10]. However, the IBVS ones need to care of the feature leaving the field of view, and they could create orbits out of ranges of the robot motion. The IBVSs only need a 3D information but with the depth information of the characteristic points. Recently, hybrid visual servoing remedies have been developed as combinations of the PBVS and IBVS ones to inherit their advantages [10, 11]. However, they have poor durability and are particularly sensitive to image noises. Moreover, it is necessary to calculate and separate the homogeneous matrix at all times during the control process, which leads to computational pressure and reduces the real-time performance of the system.

In order to solve the problem of applying the IBVS to robots, in [11], a new method was proposed to estimate the depth composition contained in the interactive matrix. However, it is necessary to know the internal parameters of the camera and employment of the Newton-Raphson method for the depth estimation. In the research [12], a new machine learning method was used for estimating the interaction matrix, however it was a complex method with a requirement of large data collection for learning. In [13], stability and robustness of the visual servoing control using the interactive matrix were discussed in which it does not need to estimate the depth composition but choosing the known value in advance. In addition, in new impressive work [14], the visual predictive controller was proposed to completely ignore the nonlinear elements of the pixel. It only considered the linear model, but the accuracy depended on the nonlinear model.

In this article, PBVS is applied for motion control of a robot arm. The camera is fixed in where it can easily observe desired positions and overarch the movement of the robot. This method also known as the eye-to-hand (fixed-eye) systems. The key feature in the article is that, we only use one camera but still being able to control the robot to the desired location in 3D space. With this method, we use a bar that has a fixed length. By relying on the information of distance error, it could determine the z-axis for the robot's movement. In addition, this control method can eliminate backlashes of the robot's joints. Effectiveness of the proposed controller is verified by comparative simulation results.

Visual servoing control is based on image processing techniques and control theories. Since there is much content, we only focus on analyzing aspects of image processing and things related to visual servoing control. The rest of the article con-sists of the presentation of the model structure presented in Section II. Section III shows relationships between points in the real coordinates and the pixel coordinates on the camera frame. The control algorithm based on the error of pixels is described in Section IV. The results of the evaluation and verification of the above method are carried out in Section V. Finally, the summary of the article is concluded in Section VI.

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II. PROBLEM STATEMENT

A robotic arm consists of a set of links that are connected together in the form of a consecutive chain. In this paper, we use a model consisting of a 3-degree-of-freedom (3-DOF) robot and a fixed camera that covers the robot's workspace. The structure of the overall system is illustrated in Fig. 1.

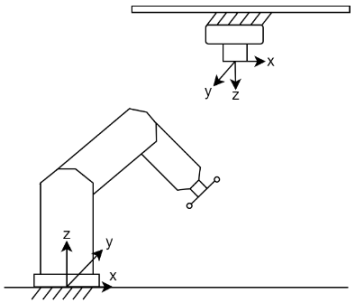


Fig. 1. The robot's structure and camera placement

With this structure, we will use a spherical that has a radius of L attached to the end-effector of the robot. The camera is used to observe the bar and collect the pixel value on the frame camera. So, we are able to control the virtual end-effector of the robot reach to the desired position. In the next parts of the paper, we design a control framework to realize the aforementioned control mission. From there, we will analyze the advantages and disadvantages of this study.

III. THE RELATIONSHIP BETWEEN ACTUAL POINTS AND CAMERA PIXELS

For a 3D point of ${}^{w}P=(X_w,Y_w,Z_w)$ in the world coordinate, we can compute position of this point relative to a camera coordinate ${}^{c}P=(X_c,Y_c,Z_c)$ using homogenous transformation:

$$ {}^{c}P = {}^{c}R_w \, {}^{w}P + {}^{c}T_w \tag{1} $$

We project ${}^{c}P$ onto the principle plane of the camera and obtain a 2D coordinate position of $p=(x,y)$:

$$ \begin{cases} x = f_x \dfrac{X_c}{Z_c} \\ y = f_y \dfrac{Y_c}{Z_c} \end{cases} \tag{2} $$

We can consider that the focal lengths in the x and y directions of the camera are the same $f = f_x = f_y$. Fig. 2 demonstrates the pixel point $p=(x,y)$ in the 2D value after using the projection on the image plane of the camera.

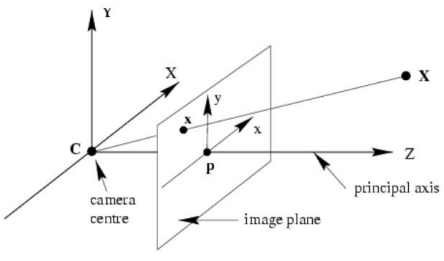


Fig. 2. The 3D point projected onto the camera frame

IV. VISUAL SERVOING CONTROL

The traditional image-based control schemes [15, 16] use specific points of images on frame camera to define a featured set s. Hence, the goal proposed in the vision-based control is to minimize a error that is determined by:

$$ e(t) = s - s^{*} \tag{3} $$

where s and s^{*} are the current and desired pixel vector, respectively. The pixel vector includes x, y, and length L values after being projected onto the camera.

The measurement error $e(t)=[e_x \quad e_y \quad e_z]^T$ on camera is obtained by three elements. Two components x and y change on the camera as the robot moves to determine the actual X, Y position. The third component is determined based on the maximum diameter of shape to estimate the depth of Z. Figure 3 shows a featured image of the desired position and a current position on the image plane of the camera. So, the error measurement is given by:

$$ \begin{cases} x_d = \dfrac{(x_{A_d} + x_{B_d})}{2}; x = \dfrac{(x_A + x_B)}{2} \\ y_d = \dfrac{(y_{A_d} + y_{B_d})}{2}; y = \dfrac{(y_A + y_B)}{2} \\ L_d = \sqrt{(x_{A_d} - x_{B_d})^2 + (y_{A_d} - y_{B_d})^2} \\ L = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2} \end{cases} \tag{4} $$

From Eq. 4, we have the error pixel as follows:

$$ \begin{cases} e_x = x_d - x \\ e_y = y_d - y \\ e_z = l - l_d \end{cases} \tag{5} $$

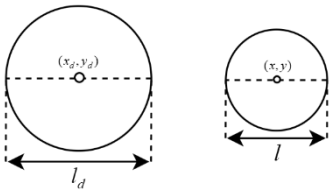


Fig. 3. The featured image in the frame camera

Since the camera is fixed, we can consider the pixel error as well as the camera speed represent by ${}^{c}v_c(t) = e(t)$. The important thing to note here is that the pixel velocity needs to transform to the end-effector point's velocity, so we have:

$$ {}^{0}v_p = \begin{bmatrix} {}^{0}R_c & [{}^{0}t_c] \times {}^{0}R_c \\ 0_{3X3} & {}^{0}R_c \end{bmatrix} {}^{c}v_c \tag{6} $$

So, we apply the Jacobian robot control method:

$$ {}^{0}v_p = J\dot{\theta} \tag{7} $$

Where, ${}^{0}v_p = [\dot{x} \quad \dot{y} \quad \dot{z}]^T$ is the vector linear velocity. J is the Jacobian vector according to θ of the robot.

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$\dot{\theta} = \begin{bmatrix} \dot{\theta}_1 & \dot{\theta}_2 & \dot{\theta}_3 \end{bmatrix}^T$ is the joint velocity vector.

By combining (6) and (7), we have the desired control velocity control signal as follows:

$$\dot{\theta} = \Gamma J^{-1\,0} v_p \tag{8}$$

Where, $\Gamma = diag(sig_1 \quad sig_2 \quad sig_3)$ is positive definite diagonal matrix that contains the corresponding learning rate constant for each joint.

The whole control process of the closed-loop system is outlined in Fig. 4. As seen in the figure, the camera is employed to provide control error signals based on pixel information collected. This error is transformed into the robot coordinate system using (6). The numerical solution (8) is adopted to result in the desired velocity control signal. The process is terminated until the error e approaches zero.

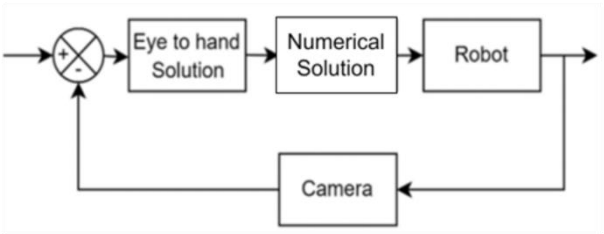


Fig. 4. The block diagram of system.

V. Simulation And Validation Result

The robot arm is considered as three rigid links connected together with lengths L_1, L_2 and L_3 respectively. This robot was able to operate in 3D space. Although this one is quite simple, it has all the essential features of a multi-degree-of-freedom robot operating in 3D space. The purpose of using a robot with 3-DOF show that the robot can still work in 3D space with one camera.

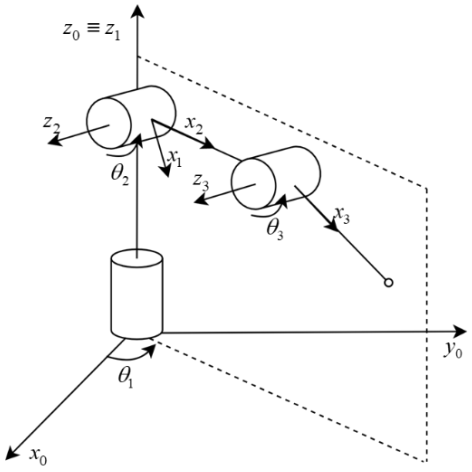


Fig. 5. The robot's structure robot 2DOF

We assume the robot has length of three joint $L_1 = 0.1(m),\ L_2 = 0.1(m),\ L_3 = 0.1(m)$. We began to calculate the relation between each of the joints through the D-H table:

TABLE I. The D-H table of 3-DOF robot

i	α	a	d	θ
1	0	0	0	θ_1
2	$\frac{\pi}{2}$	0	0	θ_2
3	0	L_2	0	θ_3

We could easily find the position end-effector of the robot:

$$\begin{cases} x = L_3 \cos\theta_1 \cos(\theta_2 + \theta_3) + L_2 \cos\theta_1 \cos\theta_2 \\ y = L_3 \sin\theta_1 \cos(\theta_2 + \theta_3) + L_2 \sin\theta_1 \cos\theta_2 \\ z = L_3 \sin(\theta_2 + \theta_3) + L_2 \sin\theta_2 \end{cases} \tag{9}$$

In order to control the robot, we apply the Jacobian numerical method (8). The choice of this method is due to the fact that visual servoing contains a velocity component for each joint. Jacobian matrix of this robot is derived as:

$$J = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_3} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_3} \\ \frac{\partial z}{\partial \theta_1} & \frac{\partial z}{\partial \theta_2} & \frac{\partial z}{\partial \theta_3} \end{bmatrix} = \begin{bmatrix} j_{11} & j_{12} & j_{13} \\ j_{21} & j_{22} & j_{23} \\ j_{31} & j_{32} & j_{33} \end{bmatrix} \tag{10}$$

where its detailed entries are as follows:

$$\begin{cases} j_{11} = -L_3 \sin\theta_1 \cos(\theta_2 + \theta_3) - L_2 \sin\theta_1 \cos\theta_2 \\ j_{12} = -L_3 \cos\theta_1 \sin(\theta_2 + \theta_3) - L_2 \cos\theta_1 \sin\theta_2 \\ j_{13} = -L_3 \cos\theta_1 \sin(\theta_2 + \theta_3) \\ j_{21} = L_3 \cos\theta_1 \cos(\theta_2 + \theta_3) + L_2 \cos\theta_1 \cos\theta_2 \\ j_{22} = -L_3 \sin\theta_1 \sin(\theta_2 + \theta_3) - L_2 \sin\theta_1 \sin\theta_2 \\ j_{13} = -L_3 \sin\theta_1 \sin(\theta_2 + \theta_3) \\ j_{31} = 0 \\ j_{32} = L_3 \cos(\theta_2 + \theta_3) + L_2 \cos\theta_2 \\ j_{33} = L_3 \cos(\theta_2 + \theta_3) \end{cases} \tag{11}$$

In the following, we discuss the control effects of using the error of pixel to control the 3-DOF robot in simulations. The structure of robot and camera placement have been shown in Fig. 1. In first simulation case, we verified the quality of the proposed control method at point P1 of (15,0,15) with initial $\begin{bmatrix} \theta_1, \theta_2, \theta_3 \end{bmatrix} = \begin{bmatrix} 0, \frac{\pi}{4}, \frac{-\pi}{2} \end{bmatrix}$ and $f = 30$. The control results obtain-ed are shown in Figs. 6-10. As seen in Figs. 6-9, with the initial position closed to the desired one, the control objective was well accomplished with a good accuracy. Figure 10 implies that the robot was controlled in a smooth manner. In second simulation case, we continue to simulate and validate the quality of the control method with a new position P2 of (10,10,5) with initial $\begin{bmatrix} \theta_1, \theta_2, \theta_3 \end{bmatrix} = \begin{bmatrix} \frac{-\pi}{6}, \frac{\pi}{4}, \frac{-\pi}{2} \end{bmatrix}$ and $f = 30$. The obtained results are depicted in Figs. 11-15. As seen in Fig. 11-12, the initial position looked far from the desired on. After applying the proposed control method, as seen in Fig. 13-14, the end-effector of the robot was driven to the goal position in a while with a very comfortable behaviors, as revealed in Fig. 15.

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In order to assess the advantages of this control method, we compare this method to the classical PBVS method [11] at the initial position of (10,-5,5) with initial $f = 30$ and $[\theta_1, \theta_2, \theta_3] = \left[\frac{\pi}{6}, \frac{\pi}{4}, \frac{-\pi}{2}\right]$ illustrated in Fig. 16. As seen in Fig. 17-18, both the methods show good steady-state control results. However, the flexible PBVS method had a better response in a smooth manner than the classical PBVS. Furthermore, the flexible PBVS method does not need to estimate the interaction matrix whereas the classical PBVS method does. The effectiveness and advantage of the proposed control method were clearly proven in these simulations.

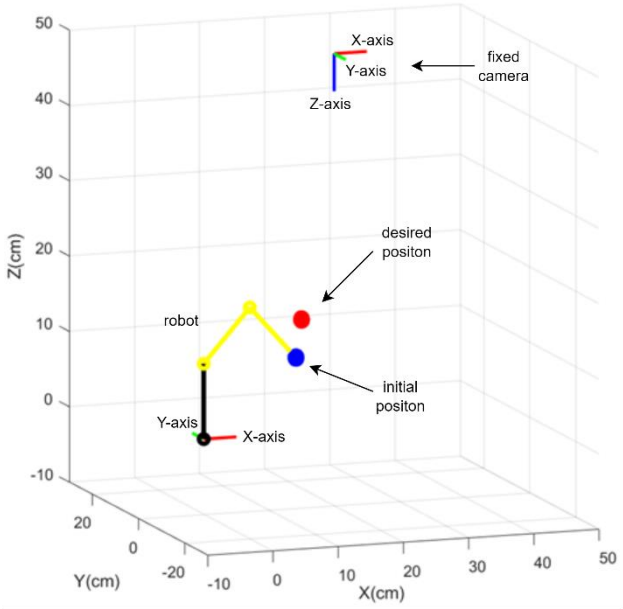


Fig. 6. The initial robot's position for first case.

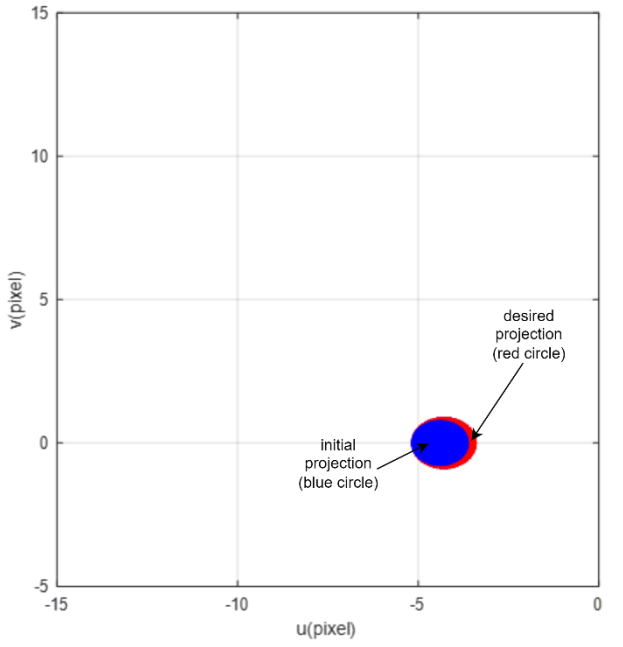


Fig. 7. The projection of a spherical on camera for first case.

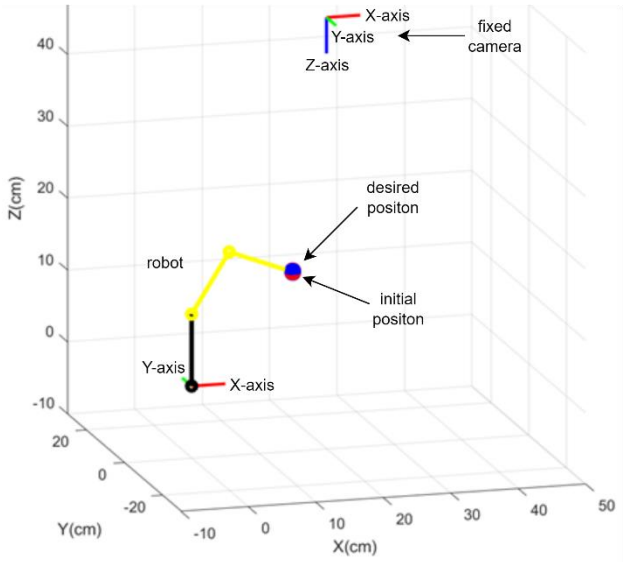


Fig. 8. The simulation result of the robot at point 3D (15,0,15).

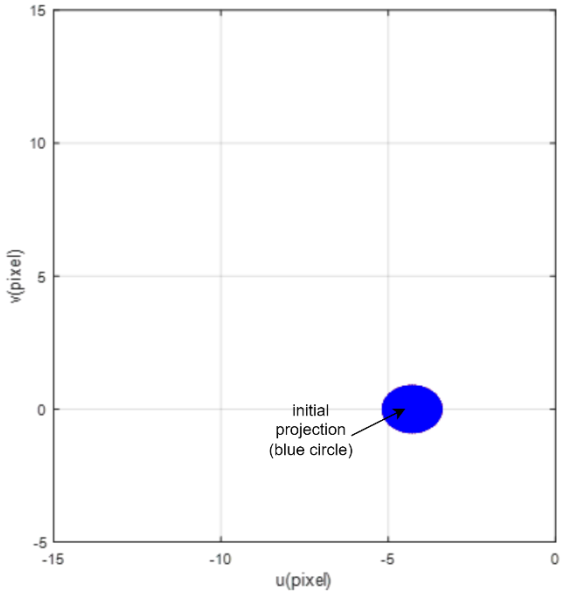


Fig. 9. The simulation result of the projection at point 3D (15,0,15).

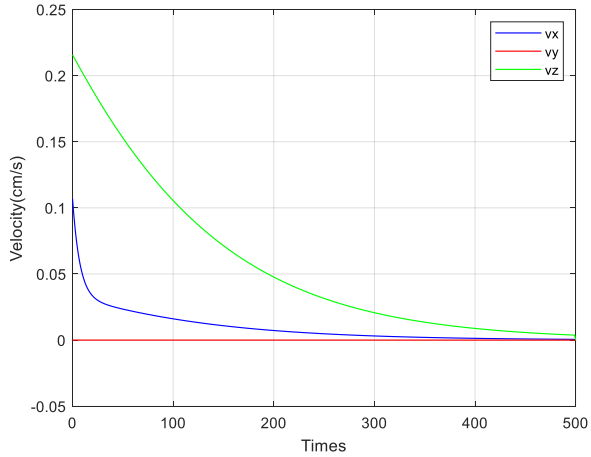


Fig. 10. The three components of position's velocity in first case

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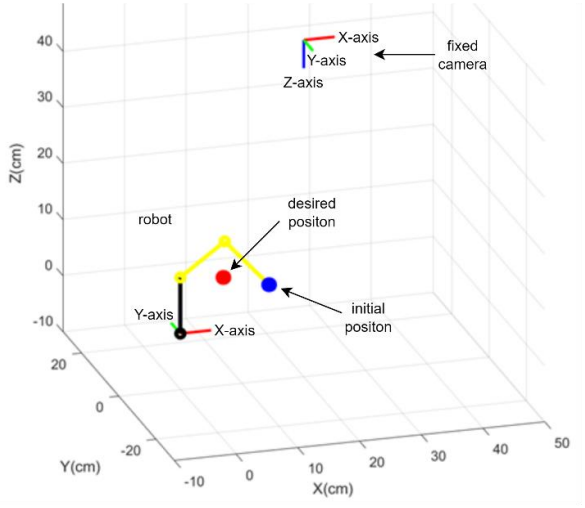


Fig. 11. The initial robot's position for second case.

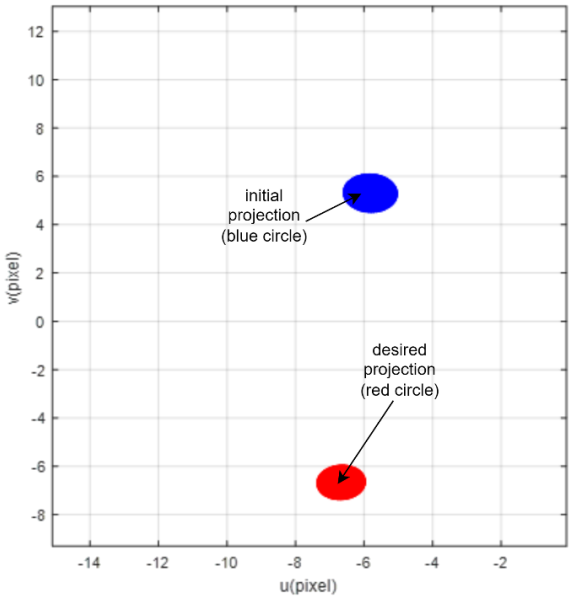


Fig. 12. The projection of a spherical on camera for second case.

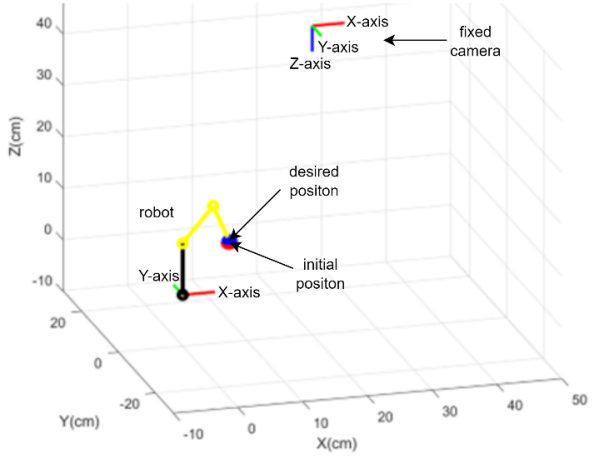


Fig. 13. The simulation result of the robot at point 3D (10,10,5).

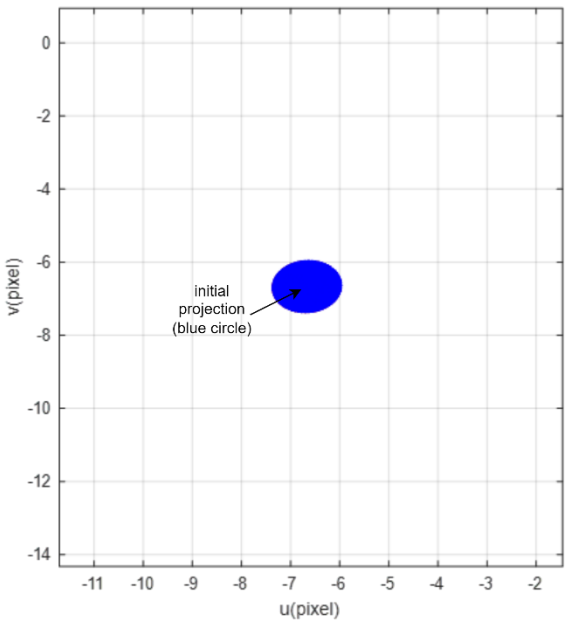


Fig. 14. The simulation result of the projection at point 3D (15,0,15).

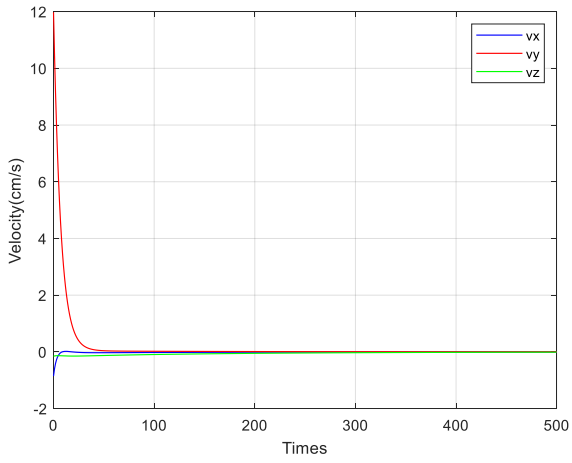


Fig. 15. The three components of position's velocity in second case.

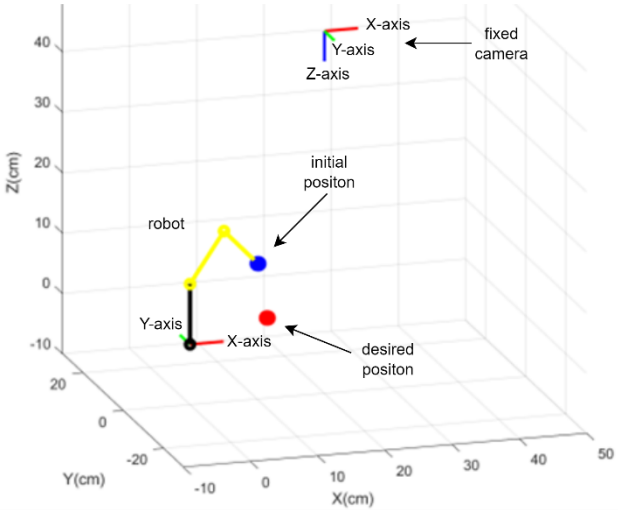


Fig. 16. The initial robot's position for compared case.

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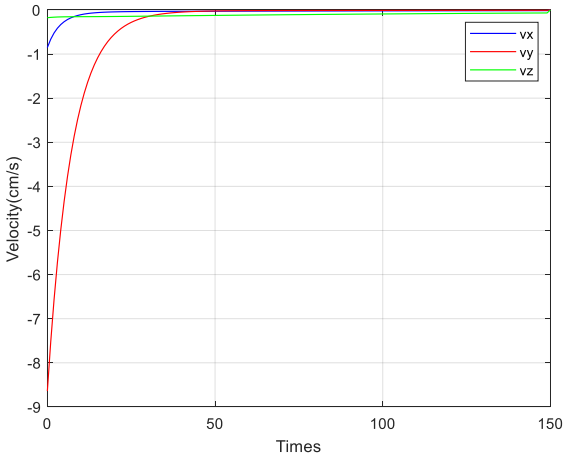


Fig. 17. The position velocity of the flexible PBVS method

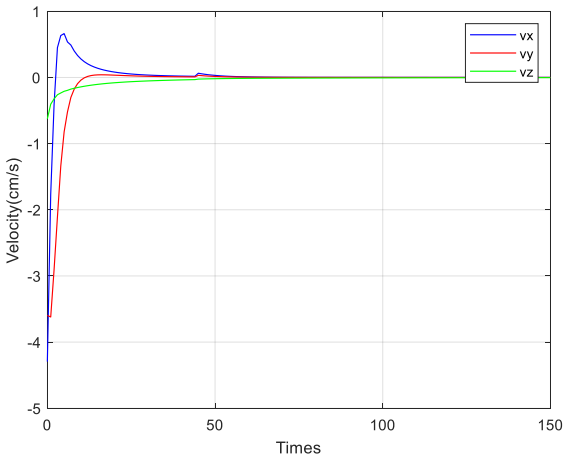


Fig. 18. The position velocity of the classical PBVS method

VI. CONCLUSIONS

In this paper, a new control method for the robot has been proposed for motion control of robotic systems. This control scheme uses a fixed camera to observe the robot's operation. So, the error based on pixels collected from the camera gives a signal to control the robot. The special feature of this technique is that it ignores the depth Z estimation of the desired position in space. The results of discussion and simulation have proved the effectiveness of this method. However, it has some limitations for the stability has not been proven.

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